

● HARD

● GEOMETRY AND TRIGONOMETRY

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# Circles

10 questions · ~15 min

# 03

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SAT QUESTION BANK

2026-05-29

Q1

GEOMETRY AND TRIGONOMETRY

Which of the following equations represents a circle in the  $xy$ -plane that intersects the  $y$ -axis at exactly one point?

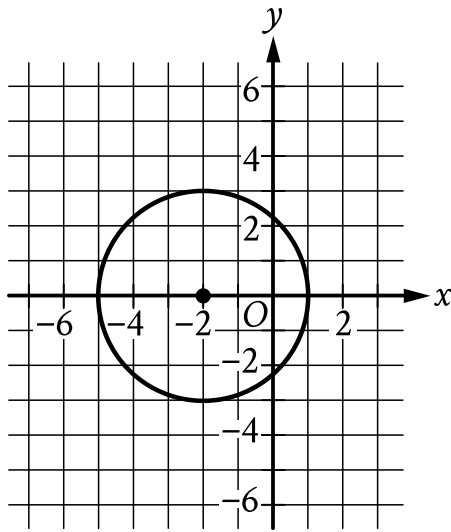
- A.  $x - 8^2 + y - 8^2 = 16$
- B.  $x - 8^2 + y - 4^2 = 16$
- C.  $x - 4^2 + y - 9^2 = 16$
- D.  $x^2 + y - 9^2 = 16$

Q2

GEOMETRY AND TRIGONOMETRY

In the  $xy$ -plane, the graph of  $2x^2 - 6x + 2y^2 + 2y = 45$  is a circle. What is the radius of the circle?

- A. 5
- B. 6.5
- C.  $\sqrt{40}$
- D.  $\sqrt{50}$



- The center of the circle is the point (negative 2 comma 0).
- Clockwise from top, the circle passes through the following points:
  - (negative 2 comma 3)
  - (1 comma 0)
  - (negative 2 comma negative 3)
  - (negative 5 comma 0)

Circle  $A$  (shown) is defined by the equation  $x + 2^2 + y^2 = 9$ . Circle  $B$  (not shown) is the result of shifting circle  $A$  down 6 units and increasing the radius so that the radius of circle  $B$  is 2 times the radius of circle  $A$ . Which equation defines circle  $B$ ?

- A.  $x + 2^2 + y + 6^2 = 49$
- B.  $2x + 2^2 + 2y + 6^2 = 9$
- C.  $x + 2^2 + y - 6^2 = 49$
- D.  $2x + 2^2 + 2y - 6^2 = 9$

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY

The equation  $(x+6)^2 + (y+3)^2 = 121$  defines a circle in the xy-plane. What is the radius of the circle?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

**Q5**

GEOMETRY AND TRIGONOMETRY

A circle in the  $xy$ -plane has equation  $(x+3)^2 + (y-1)^2 = 25$ . Which of the following points does NOT lie in the interior of the circle?

- A.  $(-7, 3)$
- B.  $(-3, 1)$
- C.  $(0, 0)$
- D.  $(3, 2)$

**Q6**

GEOMETRY AND TRIGONOMETRY

What is the diameter of the circle in the  $xy$ -plane with equation  $x^2 - 5^2 + y^2 - 3^2 = 16$ ?

- A. 4
- B. 8
- C. 16
- D. 32

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

Circle A in the  $xy$ -plane has the equation  $x + 5^2 + y - 5^2 = 4$ . Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the  $xy$ -plane is  $x + 5^2 + y - 5^2 = k$ , where  $k$  is a constant. What is the value of  $k$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

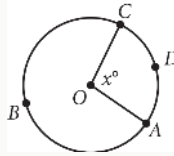
A circle in the  $xy$ -plane has its center at  $(19, 18)$  and has a radius of  $9k$ . Which equation represents this circle?

A.  $(x - 19)^2 + (y - 18)^2 = 81k$

B.  $(x - 19)^2 + (y - 18)^2 = 81k^2$

C.  $(x - 19)^2 + (y - 18)^2 = 9k$

D.  $(x - 19)^2 + (y - 18)^2 = 9k^2$



The circle above has center O, the length of arc  $\widehat{ADC}$  is  $5\pi$ , and  $x = 100$ . What is the length of arc  $\widehat{ABC}$  ?

- A.  $9\pi$
- B.  $13\pi$
- C.  $18\pi$
- D.  $\frac{13}{2}\pi$



A circle has center  $O$ , and points  $A$  and  $B$  lie on the circle. The measure of arc  $AB$  is  $45^\circ$  and the length of arc  $AB$  is 3 inches. What is the circumference, in inches, of the circle?

- A. 3
- B. 6
- C. 9
- D. 24